Name: ______________________________

WORKSHEET

Sketching periodic functions – amplitude and period

15 12 10 10 9 15 15   14 15   12 15 12 7 16 16 3

4 2 9   10 12 16 5 14 13 7 4 14 8 13   8 11

10 8 13 4 6 8 16 16 14 13 1   11 7 14 16 12 6 9

Sylvester Stallone

Rule

Given a periodic function $y = A \sin Bx$

A is the **amplitude**

$\frac{2\pi}{B}$ is the **period**

Questions

1 What is the amplitude of $y = 3 \sin (x)$?

2 Sketch this function.

3 What is the period of $y = \tan (x)$?

4 Sketch this function.

5 What is the amplitude of $y = 5 \cos (x)$?

6 Sketch this function.

7 What is the period of $y = \sin (4x)$?

8 Sketch this function.

For the function $y = 2 \cos (3x)$:

9 State the amplitude.

10 State the period.

11 Sketch this function.

For the function $y = 4 \tan \left(\frac{1}{2}x\right)$:

12 State the amplitude.

13 State the period.

14 Sketch this function.

15 Sketch the graph of $y = 3 \sin (2x)$.

16 Sketch the graph of $y = 7 \cos (4x)$.

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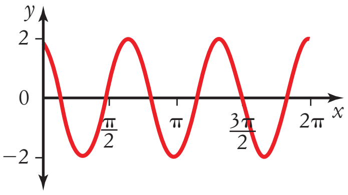
Solutions

A $\frac{\pi}{2}$

C $\frac{2\pi}{3}$

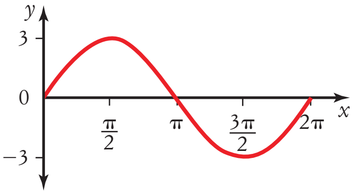
E 2

F



G 3

H



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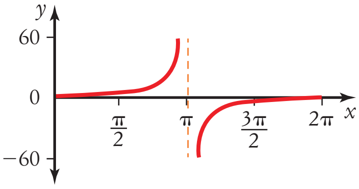

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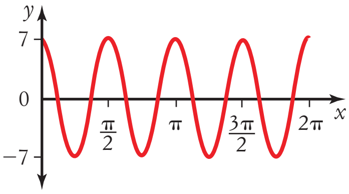
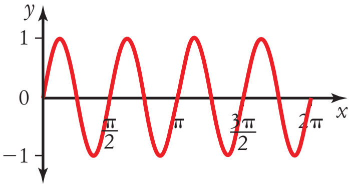


M 5

N 2π

O



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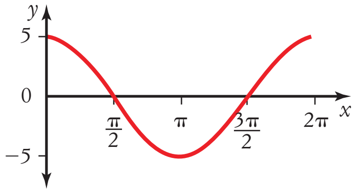

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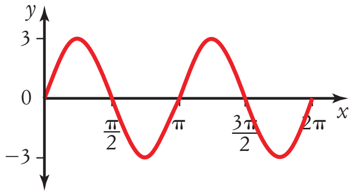

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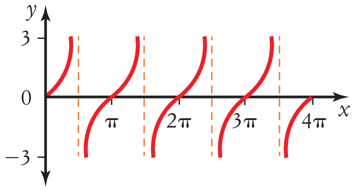

U 4

Y π